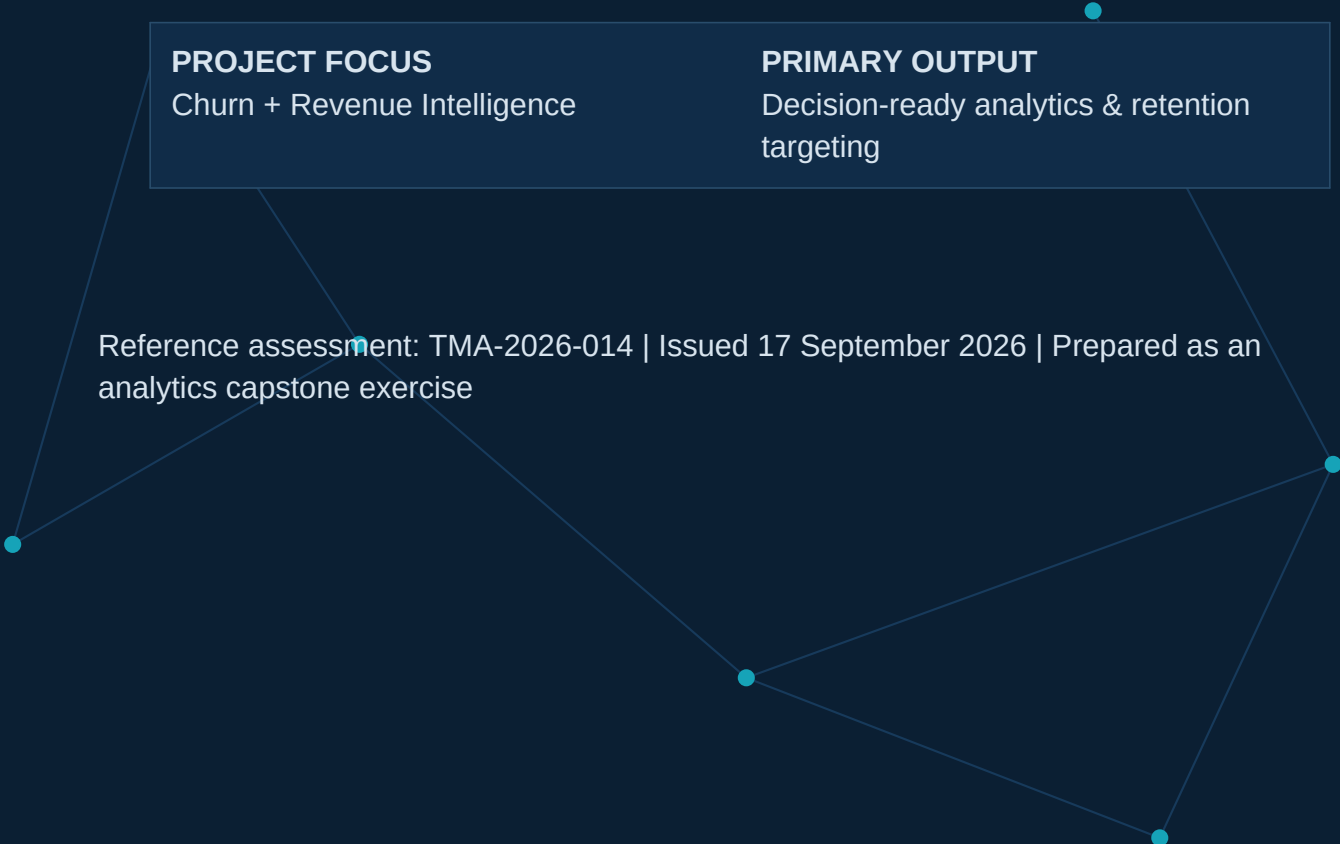




JIO SUBSCRIBER RETENTION

Business Analytics Project Report

A concise business blueprint for diagnosing churn, quantifying revenue exposure, predicting customer risk, and designing targeted retention actions.



PROJECT FOCUS

Churn + Revenue Intelligence

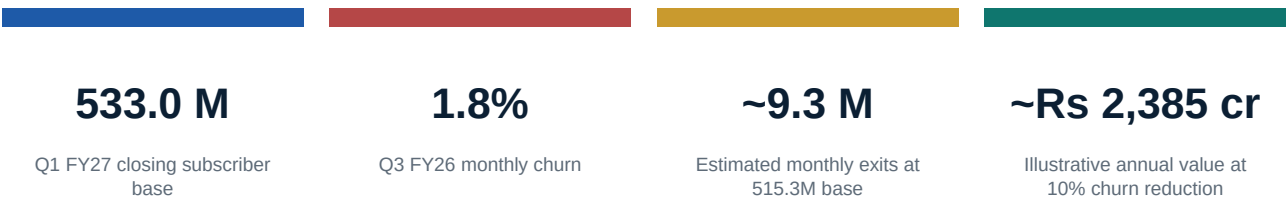
PRIMARY OUTPUT

Decision-ready analytics & retention targeting

Reference assessment: TMA-2026-014 | Issued 17 September 2026 | Prepared as an analytics capstone exercise

1. Executive Summary

The project is not simply about measuring churn. It is about turning subscriber behaviour into an actionable retention system: understand where churn is concentrated, explain the drivers, predict who is likely to leave, identify customer value at risk, and target interventions that can genuinely change outcomes.



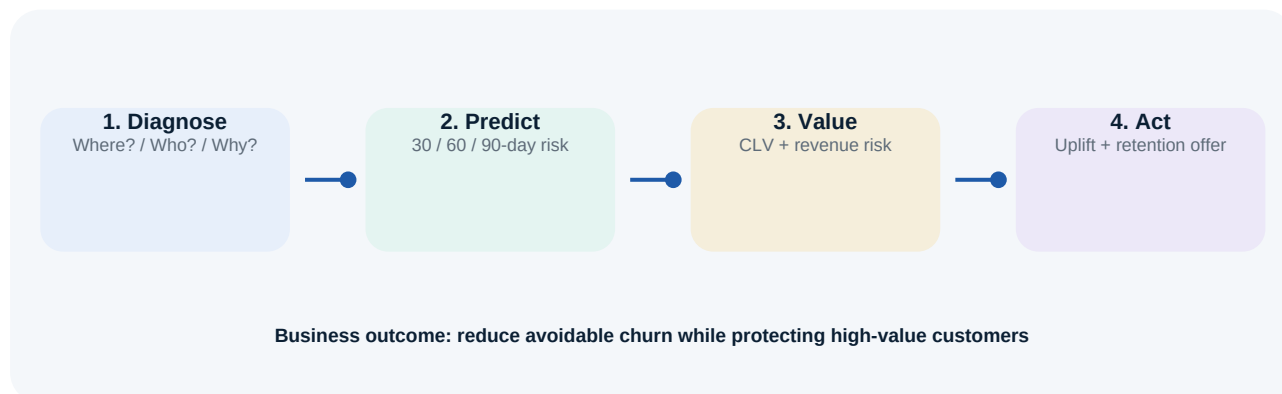
What the analysis must establish

Business question	What to find	Decision enabled
Where is churn concentrated?	Circle, ARPU, tenure, plan and cohort patterns	Prioritise geographies and segments
Why are customers leaving?	Price sensitivity, recharge gaps, service tickets, network and usage signals	Design the right intervention
Who is worth defending?	Churn probability combined with customer value / CLV	Protect high-value revenue
Who is persuadable?	Uplift / treatment response, not risk alone	Reduce wasted retention spend

Important: the Rs 23,800 crore annual exposure is an order-of-magnitude anchor, not a forecast. It applies reported ARPU uniformly and should be replaced by segment/value-weighted estimates once internal data is available.

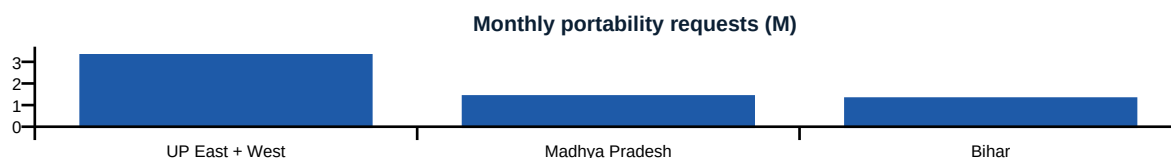
2. What the Project Needs to Find

The analytical story should move from descriptive evidence to diagnosis and then to action. The following questions define the core scope.



Analysis area	Key questions	Primary metrics / methods
Base & churn	Where and among whom is churn highest?	Churn %, exits, circle, plan, tenure, cohort
Behaviour	What changes before churn?	Recharge gaps, frequency, usage, app activity, plan migration
Service & network	Do experience failures predict exits?	Ticket ageing, reopen rate, RSRP, SINR, drops, throughput
Competitive pressure	Where is portability risk concentrated?	Port-out volume, circle heatmap, timing around tariff changes
Revenue risk	How much value is exposed?	ARPU, CLV, revenue-at-risk, high-value risk deciles
Intervention	Who is likely to respond to treatment?	Uplift, treatment/control, redemption, incremental retention

Priority market signals



The brief identifies portability as heavily concentrated in Uttar Pradesh East + West, Madhya Pradesh and Bihar; these areas are proposed as pilot territories, while the circle-level churn ranking remains directional until internal churn data is available.

3. Data, KPI and Dashboard Blueprint

The project requires a cross-functional customer 360 view. The strongest design joins CRM, billing/recharge, CDR, portability, network, care, campaigns, app behaviour and planning workbooks.

Source	Critical fields	Why it matters
CRM / Subscriber master	Customer, tenure, circle, plan, device, segment	Customer profile and segmentation
Billing / Recharge	Value, cadence, gaps, payment failures, plan history	Price sensitivity and inactivity signals
CDR / mediation	Voice, SMS, data sessions	Usage trend and engagement
Portability gateway	Port-in/out requests and outcomes	Competitive intent / direct churn signal
Network OSS	RSRP, SINR, PRB, drops, throughput	Experience-to-churn relationship
Care ticketing	Category, ageing, reopen, resolution	Service failure signals
Campaign management	Offer exposure, redemption, treatment/control	Retention economics and uplift
App clickstream	Self-care and plan-browsing behaviour	Digital intent and engagement
Planning workbooks	Targets, offers, margins, escalation rules	Commercial context outside the warehouse

Executive dashboard - recommended KPI structure

Layer	KPIs
Business	Subscriber base Monthly churn Churned subscribers ARPU Revenue at risk Retained revenue Port-outs Net adds
Customer	High-risk customers High-value high-risk customers Tenure Recharge gap Ticket rate 5G adoption
Model	Churn probability Risk decile PR-AUC Recall@Top 10% Lift CLV Uplift score

Recommended dashboard logic: Where? -> Who? -> Why? -> How much? -> What action?

4. Predictive Analytics and Retention Design

The model layer should be designed around business usefulness, not headline accuracy. With churn near 1.8%, a naive model can look accurate while being commercially useless.

Component	Purpose	Recommended approach
Churn model	Predict 30 / 60 / 90-day probability of exit or port-out	Logistic regression baseline; gradient-boosted trees as champion candidate
Model evaluation	Measure performance under strong class imbalance	PR-AUC, recall at top decile, lift, calibrated probability
Explainability	Show why a customer is high risk	SHAP feature attributions
CLV model	Estimate expected 12-month contribution	Customer-value regression / segment-aware approach
Uplift model	Identify customers who can actually be changed by treatment	Two-model or causal-forest uplift, dependent on treatment/control history

The decision engine

RISK Will they leave?	VALUE What are they worth?	RESPONSE Can treatment change the outcome?	ACTION What should we do?
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Target principle: prioritise persuadable, high-value customers rather than everyone with a high churn score.

Retention design principle: propensity answers who is likely to leave; uplift answers who is likely to stay because we intervene. The second question is what protects retention spend from being wasted on customers who would stay anyway.

5. Delivery Roadmap and Final Project Outputs

The project should be executed in a staged sequence so that data definitions and measurement are settled before predictive or generative components are scaled.

Phase	Scope	Primary output
1. Diagnose	Data validation + churn / cohort / segment / driver analysis	Executive diagnostic dashboard + findings note
2. Predict	Churn + CLV models in shadow mode	Validated risk and value scores
3. Target	Uplift modelling + controlled pilot in priority circles	Treatment strategy and incremental retention measurement
4. Assist	Conversational analytics for SQL + workbook retrieval	Faster, governed decision support
5. Scale	National rollout + adjacent opportunities	Operating retention capability

Final deliverables for the analytics project

01	Executive KPI Dashboard A management view of subscriber base, churn, ARPU, revenue risk and retention impact.
02	Churn Analysis Circle, ARPU, tenure, plan, usage, network and service-driver analysis.
03	Customer Risk Table Customer-level churn probability, risk band, CLV and top risk drivers.
04	Predictive Model 30 / 60 / 90-day churn scoring with business-focused evaluation.
05	Revenue-at-Risk Value-weighted exposure and high-value risk segmentation.
06	Retention Targeting Uplift-based intervention logic and pilot measurement.

Critical open items before modelling

- Confirm the churn denominator and exact churn definition.
- Establish whether historical treatment/control assignment exists in campaign data.
- Replace the generic revenue anchor with segment-level / value-weighted churn economics.
- Confirm whether home connectivity is part of this project or a separate workstream.
- Assign owners for modelling, uplift and rollout phases.

Bottom line: the project succeeds when analytics moves beyond reporting churn and produces a measurable, value-weighted retention decision system.

Source: Reliance Jio - Subscriber Retention and Revenue Intelligence Assessment, TMA-2026-014, v0.3 (17 Sep 2026). Figures are as-disclosed or directional as noted in the source assessment.